

Name: _____

Class: _____

Date: _____

Lab 12: Functions Factory

Python Lab · Section 3.12–3.13 Procedures



Goal

Write functions with parameters and return values, and refactor repeated code into procedures.

Warm-up: Predict

```
def double(n):  
    return n * 2  
  
print(double(double(3)))
```

Predict the output. What are the parameter and the arguments?

AP Connection

- `def name(params):` ≈ PROCEDURE `name(params)`.
- `return` ≈ RETURN — it ends the procedure.
- Your Create Task needs a procedure with a parameter!

Build It

Step 1. Write and call your first function.

```
def area(width, height):  
    return width * height  
  
print(area(3, 4))  
print(area(10, 2))
```

Step 2. Return vs. print: what's the difference? Try both.

```
def show_total(a, b):  
    print(a + b)  
  
def get_total(a, b):  
    return a + b  
  
x = get_total(2, 3) * 10    # works  
y = show_total(2, 3) * 10  # what happens?
```

Step 3. Refactor: this code repeats itself. Replace it with one function with a parameter.

```
print("Happy birthday to you")  
print("Happy birthday dear Sam")  
print("Happy birthday to you")  
print("Happy birthday dear Ava")
```

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Build It (continued) · Level Up

Build It (continued)

Step 4. Write `letter_grade(score)` that RETURNS "A", "B", "C", or "F". Use it on a list of scores.

```
def letter_grade(score):  
    # your code  
  
for s in [95, 83, 71, 40]:  
    print(s, letter_grade(s))
```

Level Up Challenges

Mild Write `is_even(n)` that returns True or False.

Medium Write `count_above(lst, limit)` that returns how many items exceed limit.

Spicy Write a function that uses sequencing, selection, AND iteration, with a parameter that changes what it does. (This is a Create Task requirement!)

Python → AP Pseudocode

```
def count_above(lst, limit):  
    count = 0  
    for x in lst:  
        if x > limit:  
            count = count + 1  
    return count
```

Write it in AP pseudocode.

Exit Ticket

Explain how a parameter makes a procedure more useful.
